

Abstract Linear Algebra

Prof. Lanier

jlancier8@uchicago.edu

Pick up a handout. Try those warm-up problems.

Next up:

- This week: Chapter 2.4-8 of *LADW*
- Quiz 3 at end of class today.
- Homework 2 is due today / tomorrow.
- Problem session today 3-5pm in Ryerson 177. (Possibly snacks.)
- ***Change:*** Open office hours on Sunday 3-4pm in **Cobb 115**
- Homework 3 distributed today, due end of next week.
- First problem swap problems due next Wednesday.

Row equivalence: matrices “connected” by row operations

Theorem: For every $A \in M_{m \times n}^{\mathbb{F}}$ there exists a unique $B \in M_{m \times n}^{\mathbb{F}}$ in RREF such that A is row equivalent B .

existence and uniqueness

(See supplement.)

existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot

$$\begin{pmatrix} * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \end{pmatrix}$$

existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot

$$\left(\begin{array}{c|cccccc} & * & * & * & * & * & * \\ & * & * & * & * & * & * \\ & * & * & * & * & * & * \\ & * & * & * & * & * & * \end{array} \right)$$

existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



A diagram illustrating a matrix structure. On the left, a solid blue L-shaped region represents the pivot area, consisting of a vertical bar and a horizontal bar. To the right of this region is a 3x6 grid of asterisks (*). A large closing curly brace } is positioned to the right of the grid, spanning all three rows.

*	*	*	*	*	*
*	*	*	*	*	*
*	*	*	*	*	*

existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot


$$\left(\begin{array}{c|cccc} & & & & \\ & & & & \\ & & & & \\ & & & & \\ \hline & * & * & * & * \\ & * & * & * & * \end{array} \right)$$

existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



existence: apply a row reduction algorithm

to REF:

either zero column, or make first non-zero entry in a column into a pivot



to RREF:

scale pivot rows to make pivots 1, make zeroes above pivots

uniqueness: base case

Prop: Every $m \times 1$ matrix is row equivalent to exactly one of

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}$$

These are the only $m \times 1$ matrices in RREF.

uniqueness: induction step on n , assume uniqueness of RREF for $m \times (n - 1)$ matrices

Suppose A is equivalent to two different matrices B, C in RREF.

$$\begin{array}{c} B \\ \left(\begin{array}{cccccc} * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \end{array} \right) \end{array} \longleftrightarrow A \longleftrightarrow \begin{array}{c} C \\ \left(\begin{array}{cccccc} @ & @ & @ & @ & @ & @ & @ \\ @ & @ & @ & @ & @ & @ & @ \\ @ & @ & @ & @ & @ & @ & @ \\ @ & @ & @ & @ & @ & @ & @ \end{array} \right) \end{array}$$

uniqueness: induction step on n , assume uniqueness of RREF for $m \times (n - 1)$ matrices

Suppose A is equivalent to two different matrices B, C in RREF.

$$\begin{array}{c} B \\ \left(\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \begin{array}{c} \text{green bar} \end{array} \right) \rightleftharpoons A \begin{array}{c} \text{green bar} \end{array} \rightleftharpoons \begin{array}{c} C \\ \left(\begin{array}{cccccc} @ & @ & @ & @ & @ & @ \\ @ & @ & @ & @ & @ & @ \\ @ & @ & @ & @ & @ & @ \\ @ & @ & @ & @ & @ & @ \end{array} \begin{array}{c} \text{green bar} \end{array} \right) \end{array}$$

uniqueness: induction step on n , assume uniqueness of RREF for $m \times (n - 1)$ matrices

Suppose A is equivalent to two different matrices B, C in RREF.

$$\begin{array}{c} B \\ \left(\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \begin{array}{c} \text{green bar} \end{array} \right) \end{array} \longleftrightarrow A \begin{array}{c} \text{green bar} \end{array} \longleftrightarrow \begin{array}{c} C \\ \left(\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \begin{array}{c} \text{green bar} \end{array} \right)
 \end{array}$$

uniqueness: induction step on n , assume uniqueness of RREF for $m \times (n - 1)$ matrices

Suppose A is equivalent to two different matrices B, C in RREF.

$$\begin{array}{c} B \\ \left(\begin{array}{cccccc} * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \end{array} \right) \left(\begin{array}{c} - \\ - \\ - \\ - \\ - \\ - \\ - \end{array} \right) \\ = \left(\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array} \right) \end{array} \longleftrightarrow A \longleftrightarrow \begin{array}{c} C \\ \left(\begin{array}{cccccc} * & * & * & * & * & * & @ \\ * & * & * & * & * & * & @ \\ * & * & * & * & * & * & @ \\ * & * & * & * & * & * & @ \end{array} \right) \left(\begin{array}{c} - \\ - \\ - \\ - \\ - \\ - \\ - \end{array} \right) \\ = \left(\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array} \right) \end{array}$$

Row equivalent matrices
have the same nullspace.

uniqueness: induction step on n , assume uniqueness of RREF for $m \times (n - 1)$ matrices

Suppose A is equivalent to two different matrices B, C in RREF.

$$\begin{array}{c}
 \begin{matrix} B \\ \left(\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \right) \end{matrix} \begin{matrix} \left(\begin{array}{c} - \\ - \\ - \\ - \\ - \\ - \\ - \end{array} \right) \\ = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \end{matrix} \\
 \leftrightarrow A \leftrightarrow \begin{matrix} C \\ \left(\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \right) \end{matrix} \begin{matrix} \left(\begin{array}{c} @ \\ @ \\ @ \\ @ \end{array} \right) \left(\begin{array}{c} - \\ - \\ - \\ - \\ - \\ - \\ - \end{array} \right) \\ = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \end{matrix} \\
 \begin{matrix} B - C \\ \left(\begin{array}{cccccc} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{matrix} \begin{matrix} \left(\begin{array}{c} - \\ - \\ - \\ - \\ - \\ - \\ - \end{array} \right) \\ = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \end{matrix}
 \end{array}$$

Row equivalent matrices have the same nullspace.

if non-zero...

this must be 0.

Row equivalent matrices have the same nullspace.

uniqueness: induction step on n , assume uniqueness of RREF for $m \times (n - 1)$ matrices

Suppose A is equivalent to two different matrices B, C in RREF.

$$B$$
$$\begin{pmatrix} * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \\ * & * & * & * & * & * & * \end{pmatrix} \begin{pmatrix} - \\ - \\ - \\ - \\ - \\ - \\ - \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Row equivalent matrices
have the same nullspace.

$$C$$
$$\begin{pmatrix} * & * & * & * & * & * & @ \\ * & * & * & * & * & * & @ \\ * & * & * & * & * & * & @ \\ * & * & * & * & * & * & @ \end{pmatrix} \begin{pmatrix} - \\ - \\ - \\ - \\ - \\ - \\ - \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$B - C$$
$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & * - @ \\ 0 & 0 & 0 & 0 & 0 & 0 & * - @ \\ 0 & 0 & 0 & 0 & 0 & 0 & * - @ \\ 0 & 0 & 0 & 0 & 0 & 0 & * - @ \end{pmatrix} \begin{pmatrix} - \\ - \\ - \\ - \\ - \\ - \\ - \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

if non-zero...

this must be 0.

n th column forced to be a pivot column; forced where pivot must be. $\Rightarrow B = C$ ■

Eventually want to show:

Every vector space with a finite spanning set has a (finite) basis.

Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

(Upshot: we lose no generality by considering the vector spaces $\mathbb{F}^n$.)

Eventually want to show:



Every vector space with a finite spanning set has a (finite) basis.

T
o
d
a
y:

Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

(Upshot: we lose no generality by considering the vector spaces $\mathbb{F}^n$.)

(Prop 2.3.3) Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

(Prop 2.3.3) Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Pf. Suppose there exist two bases M and N , of size m and n , with $m \geq n$.

(Prop 2.3.3) Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Pf. Suppose there exist two bases M and N , of size m and n , with $m \geq n$.

Suppose $m > n$. Set up coordinates in terms of N ,
and consider linear combinations of M equaling the zero vector.

(Prop 2.3.3) Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Pf. Suppose there exist two bases M and N , of size m and n , with $m \geq n$.

Suppose $m > n$. Set up coordinates in terms of N ,
and consider linear combinations of M equaling the zero vector.

This yields an $n \times m$ matrix, which can't have a pivot in every column.
So there is a free variable, and the system has a nontrivial solution.
So M is not linearly independent.

Therefore $m = n$. ■

(Prop 2.3.3) Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Pf. Suppose there exist two bases M and N , of size m and n , with $m \geq n$.

Suppose $m > n$. Set up coordinates in terms of N ,
and consider linear combinations of M equaling the zero vector.

This yields an $n \times m$ matrix, which can't have a pivot in every column.
So there is a free variable, and the system has a nontrivial solution.
So M is not linearly independent.

Therefore $m = n$. ■

Note: we are using row reduction!

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

Pf. Let V be a vector space of finite dimension n .

Choose a basis for V : $\{v_1, \dots, v_n\}$.

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

Pf. Let V be a vector space of finite dimension n .

Choose a basis for V : $\{v_1, \dots, v_n\}$.

Consider the linear transformation $T : V \rightarrow \mathbb{F}^n$ that sends v_i to the i th standard basis vector.

It suffices to show that T is injective and surjective.

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

Pf. Let V be a vector space of finite dimension n .

Choose a basis for V : $\{v_1, \dots, v_n\}$.

Consider the linear transformation $T : V \rightarrow \mathbb{F}^n$ that sends v_i to the i th standard basis vector.

It suffices to show that T is injective and surjective.

Injective: T is injective if the only vector that maps to the zero n -tuple in $\mathbb{F}^n$ is the zero vector of V . But all other vectors in V are non-trivial linear combinations of the v_i and so do not map to the zero n -tuple in $\mathbb{F}^n$.

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

Pf. Let V be a vector space of finite dimension n .

Choose a basis for V : $\{v_1, \dots, v_n\}$.

Consider the linear transformation $T : V \rightarrow \mathbb{F}^n$ that sends v_i to the i th standard basis vector.

It suffices to show that T is injective and surjective.

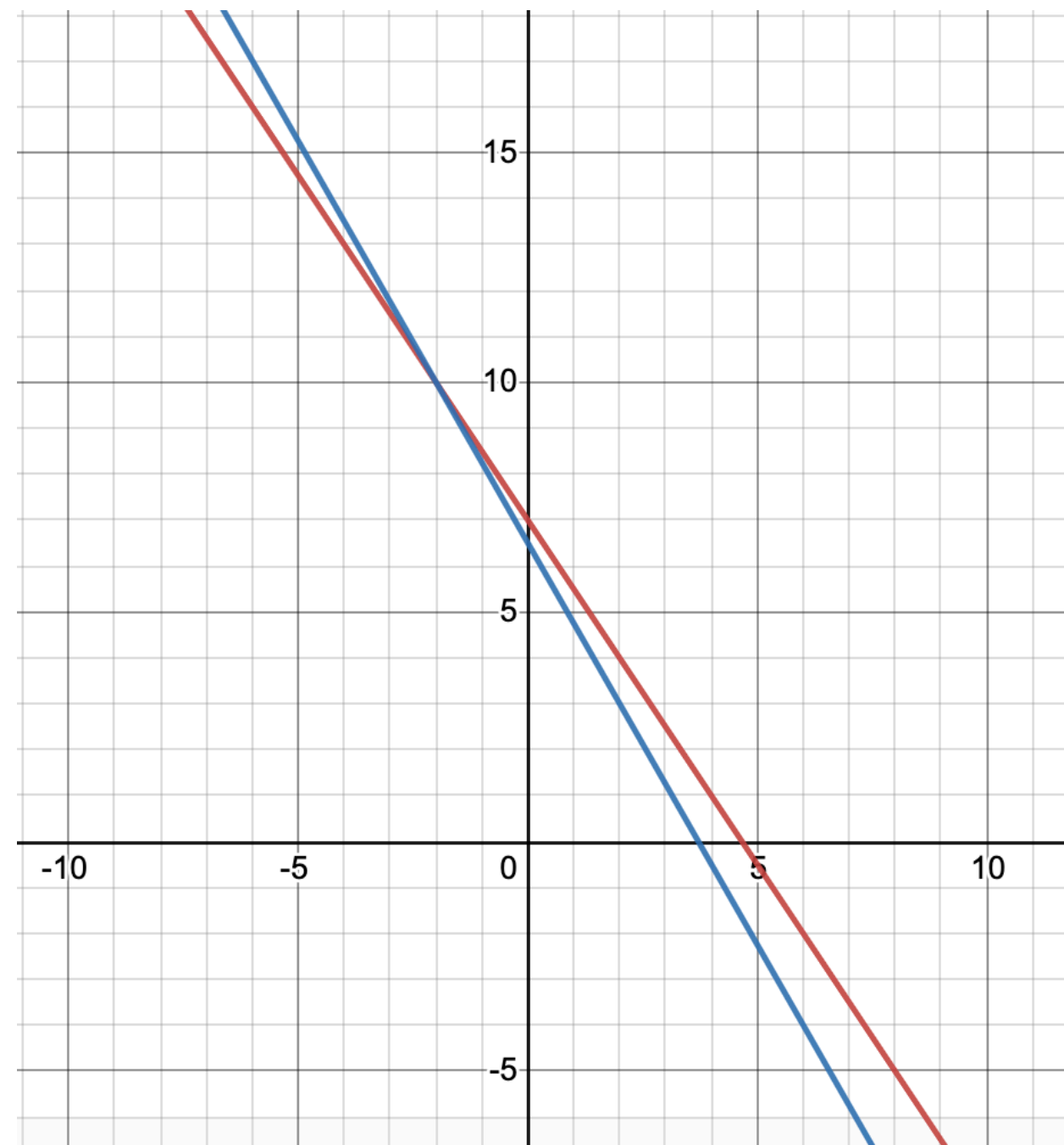
Injective: T is injective if the only vector that maps to the zero n -tuple in $\mathbb{F}^n$ is the zero vector of V . But all other vectors in V are non-trivial linear combinations of the v_i and so do not map to the zero n -tuple in $\mathbb{F}^n$.

Surjective: for any element of $\mathbb{F}^n$, it is hit by a linear combination v of the v_i —the coordinates give the recipe to form the appropriate v . ■

system of linear equations

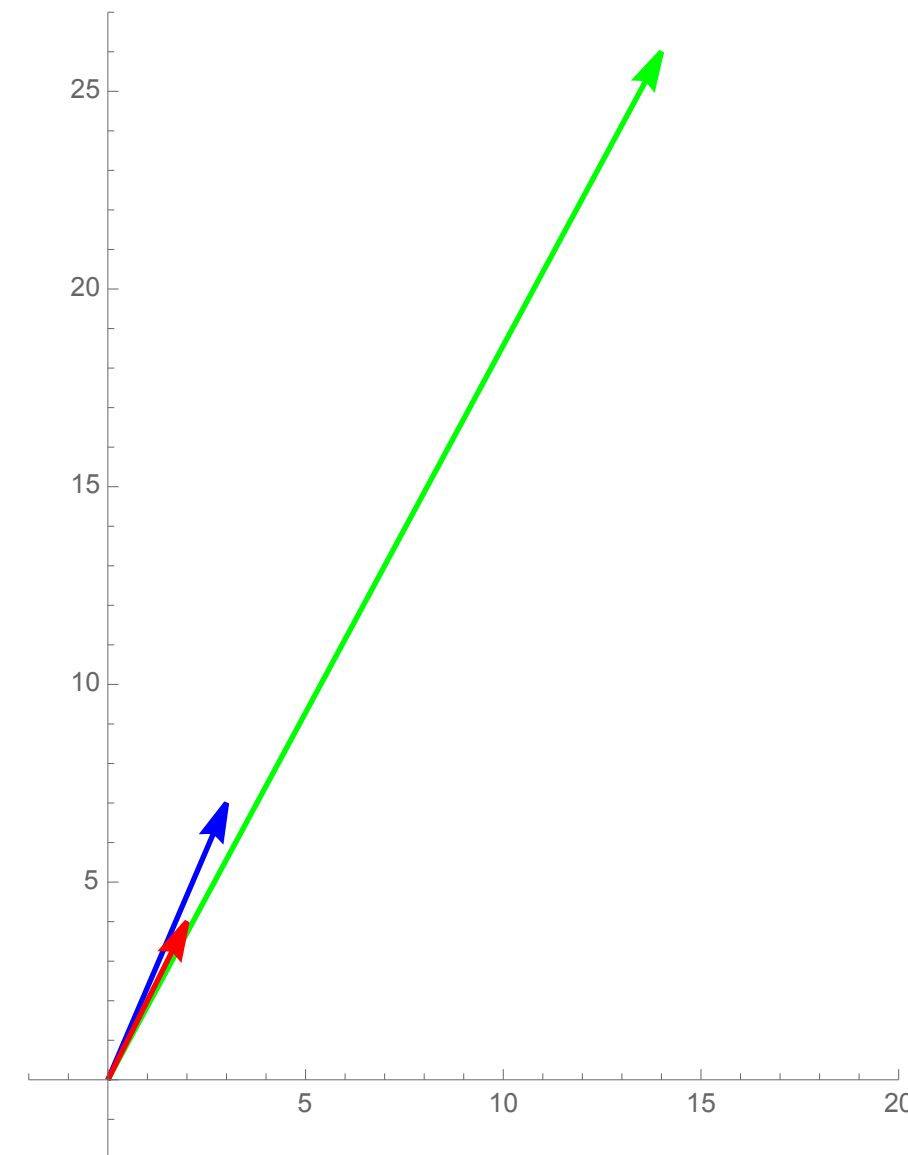
$$3x + 2y = 14$$

$$7x + 4y = 26$$



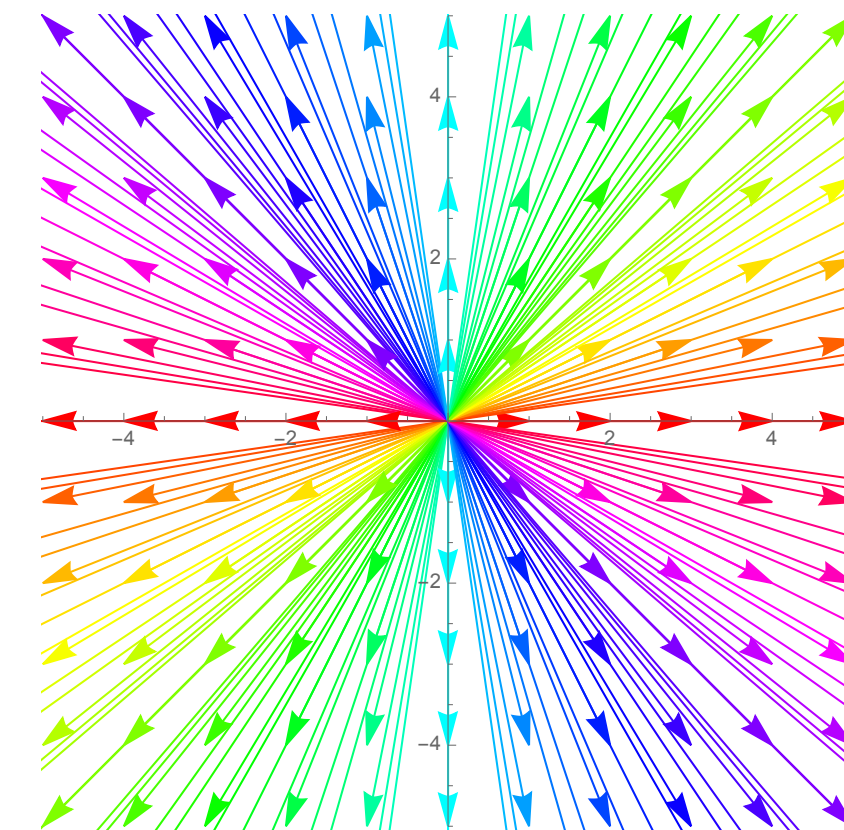
vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$

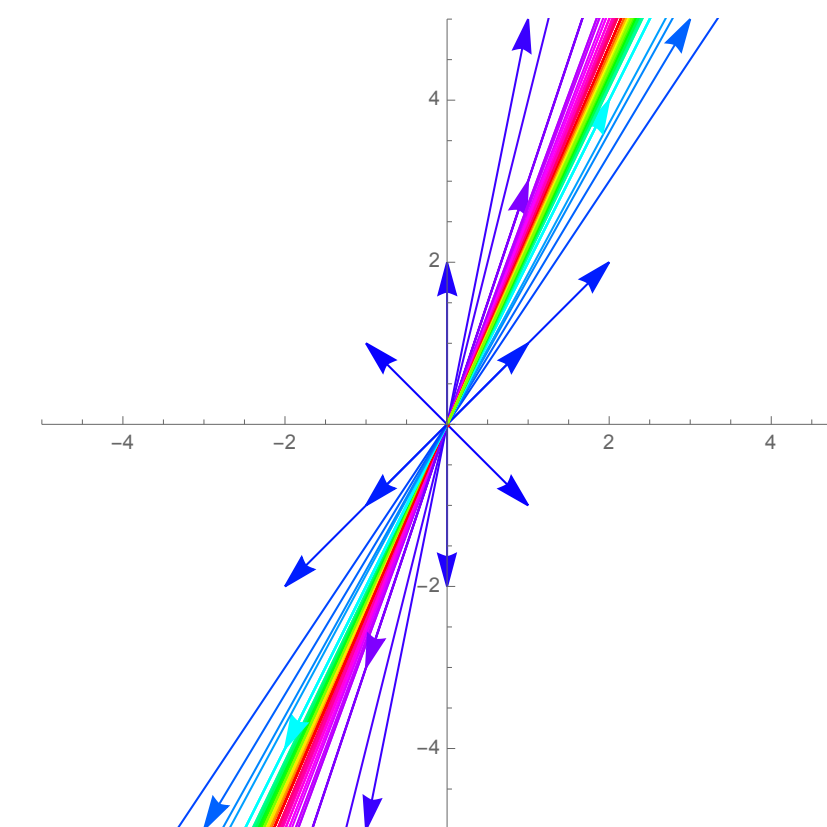


matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$



$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$$

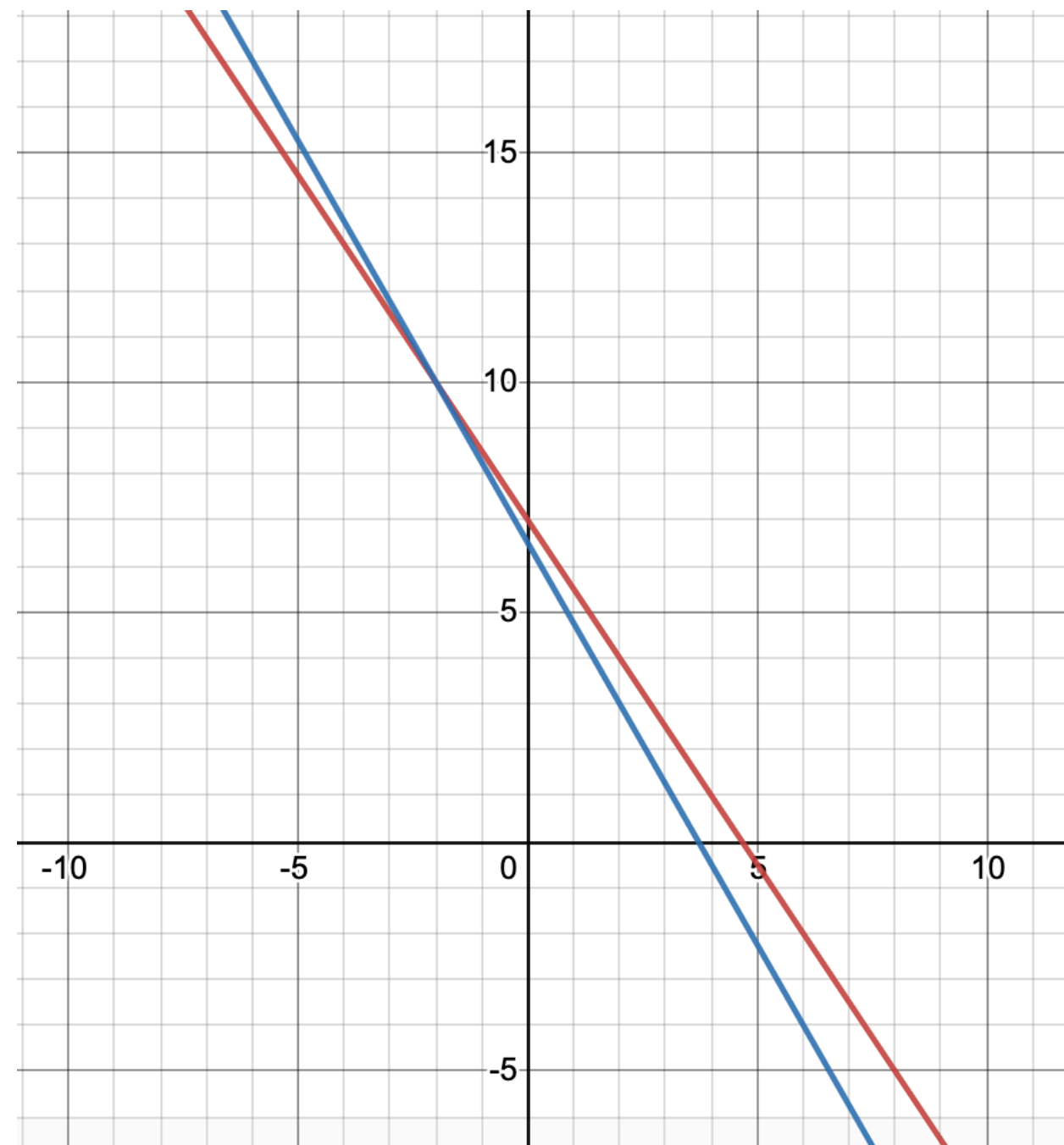


When does a solution exist?
When is the solution unique?

system of linear equations

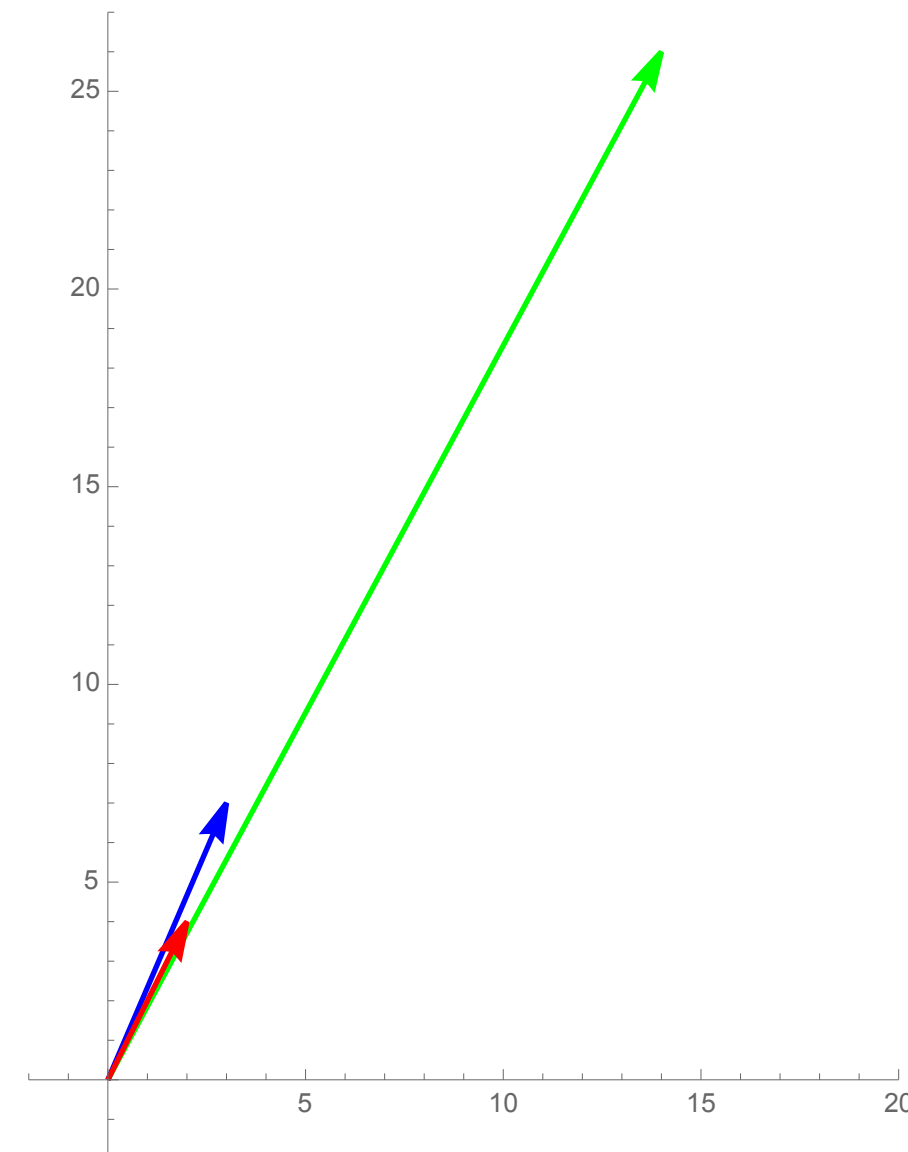
$$3x + 2y = 14$$

$$7x + 4y = 26$$



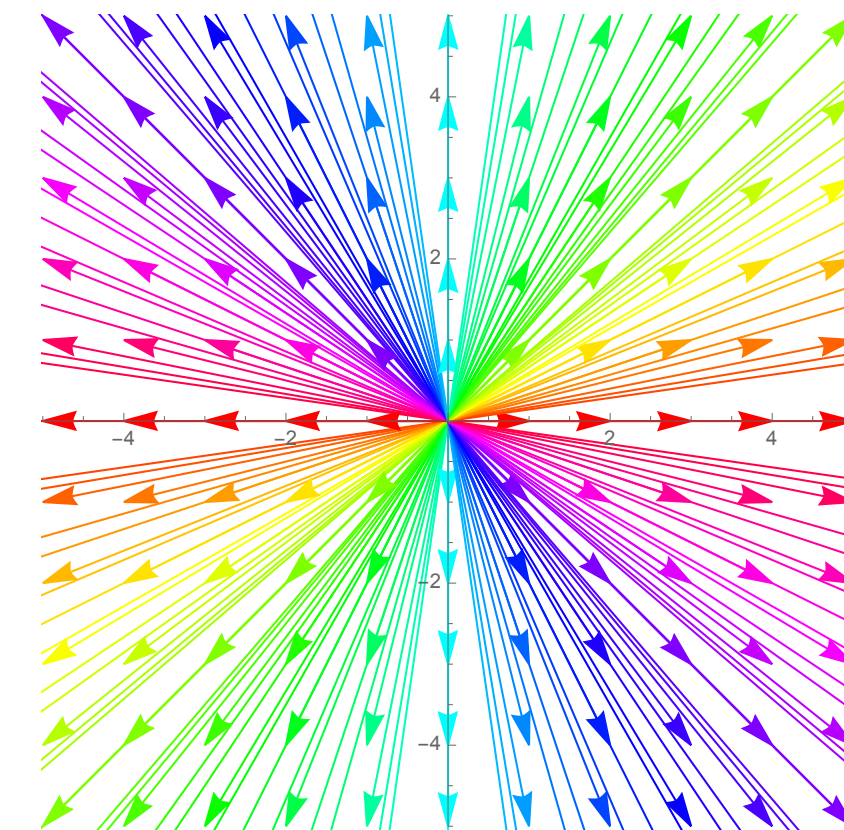
vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$

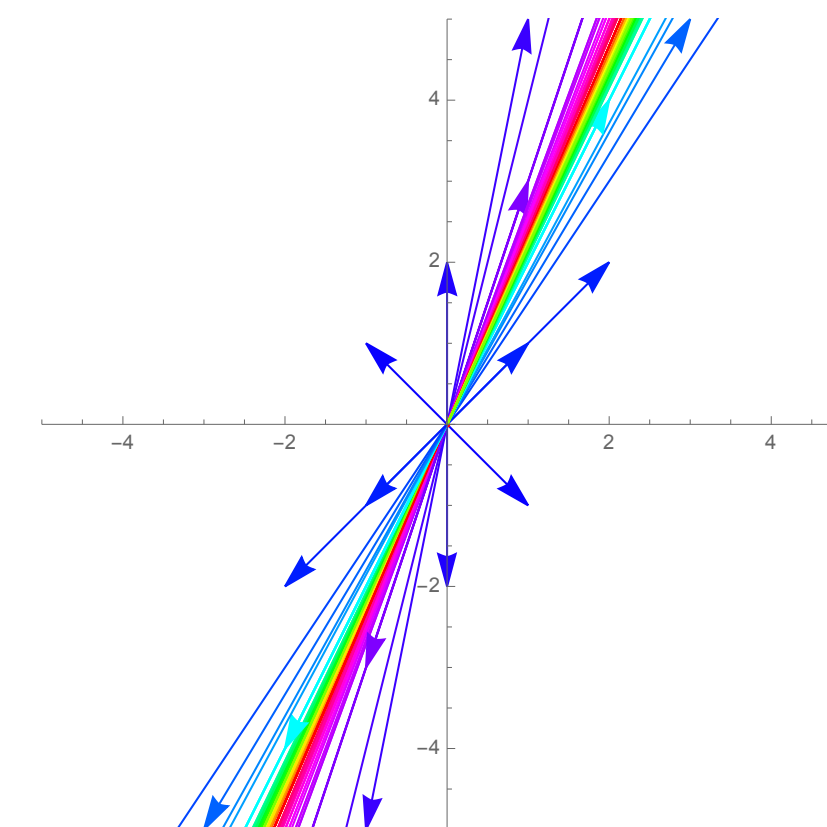


matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$



$$\downarrow \begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$$

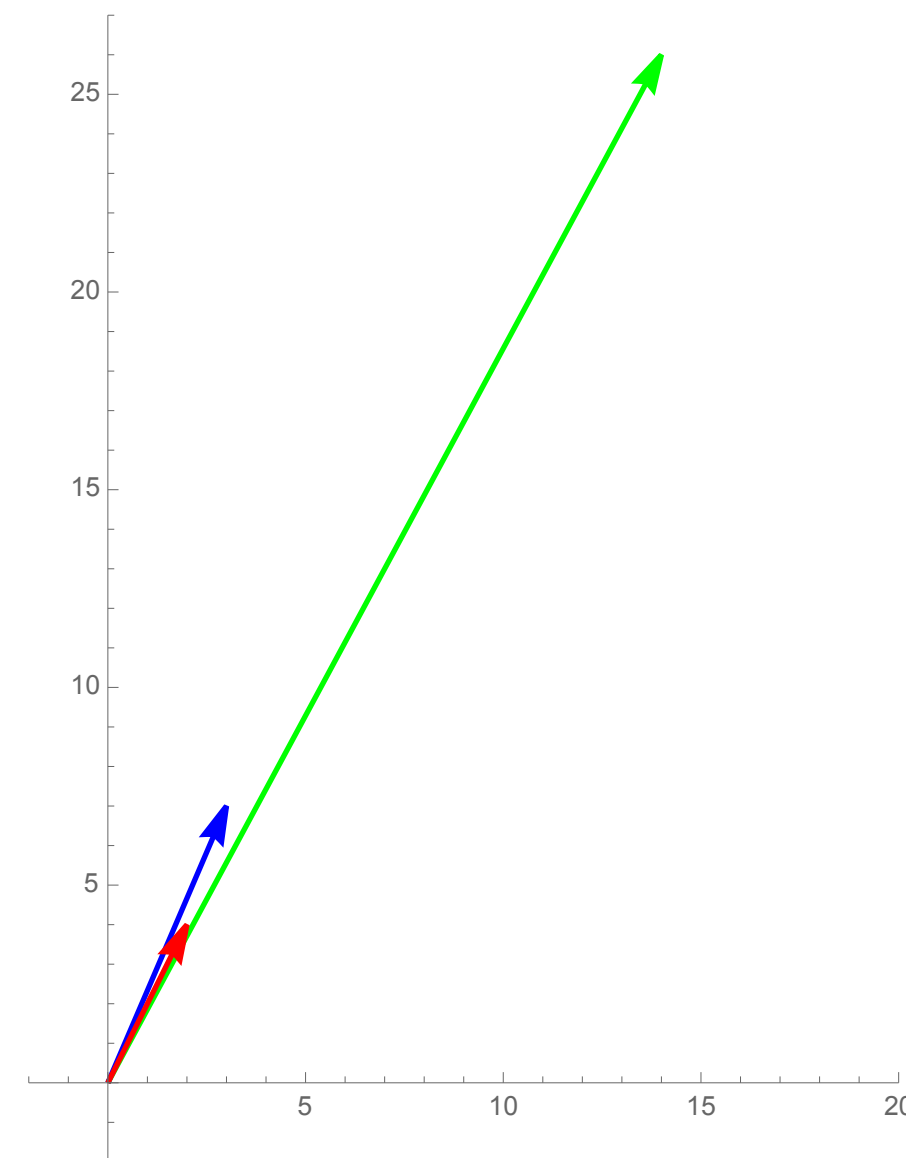


When does a solution exist?
When is the solution unique?

We can decide
by row reduction.

vector equation

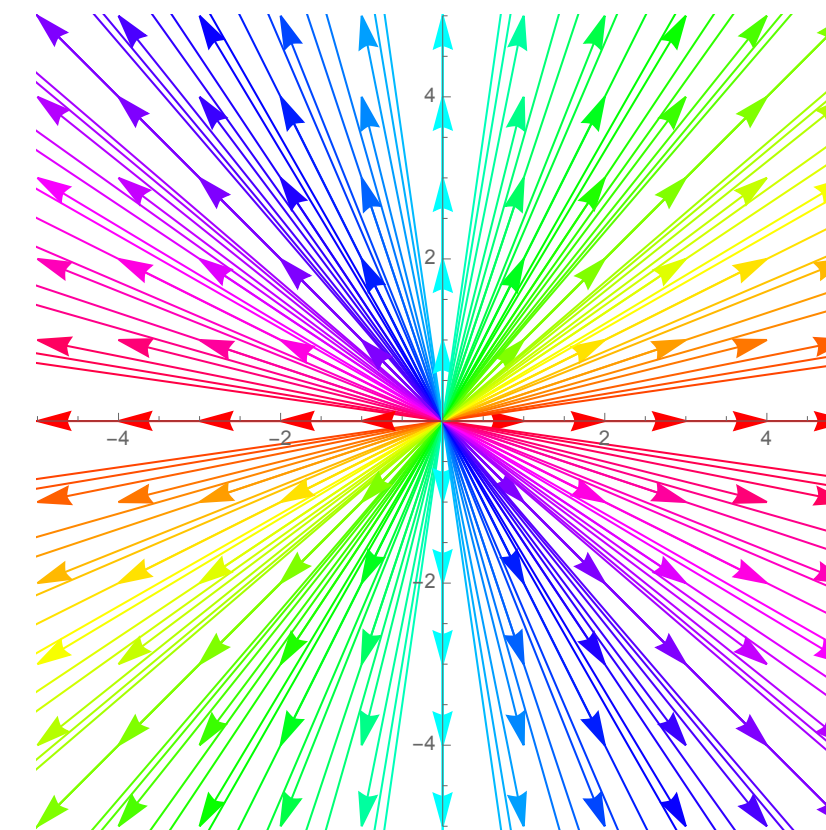
$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \text{?}$$



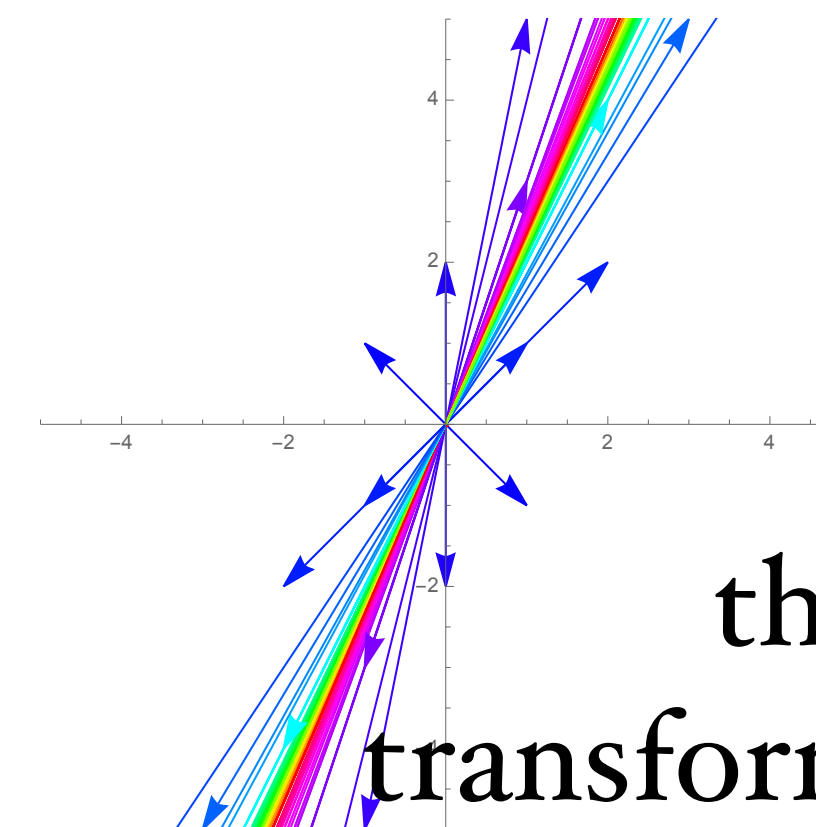
the vectors span
the vector space

matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \text{?}$$



$\downarrow \begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$

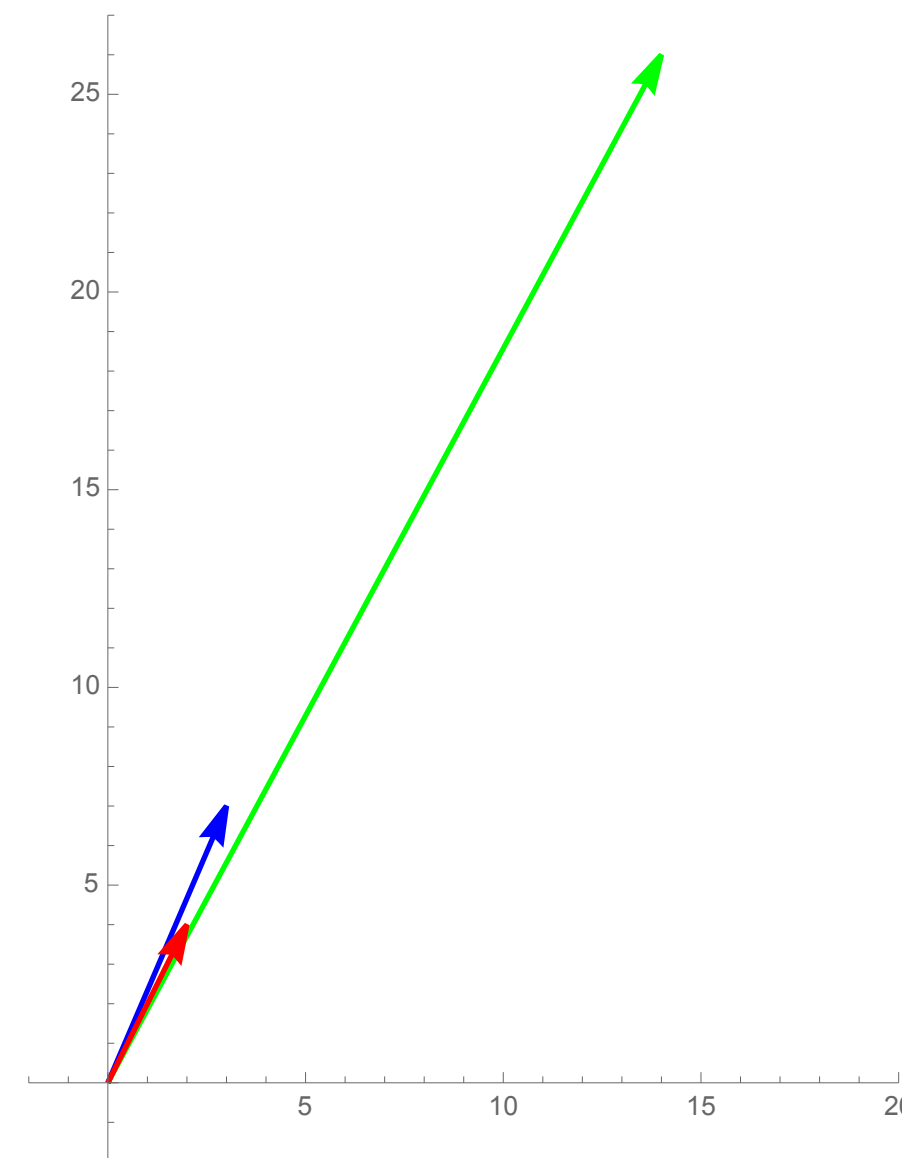


the linear
transformation is onto

Solution *always* exists if:

vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \text{?}$$

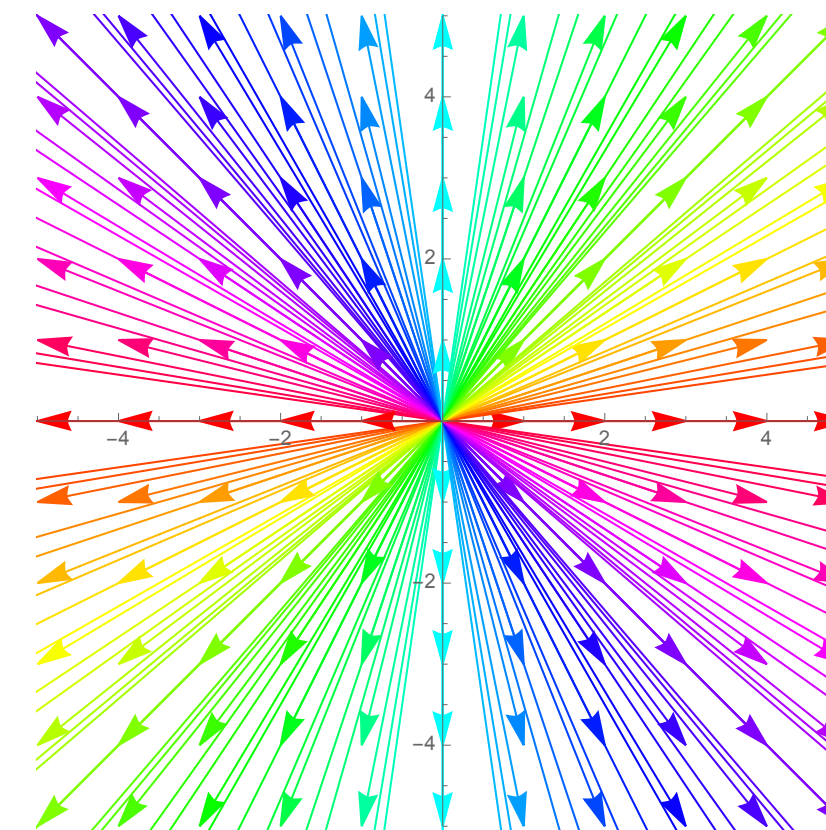


the vectors are linearly independent

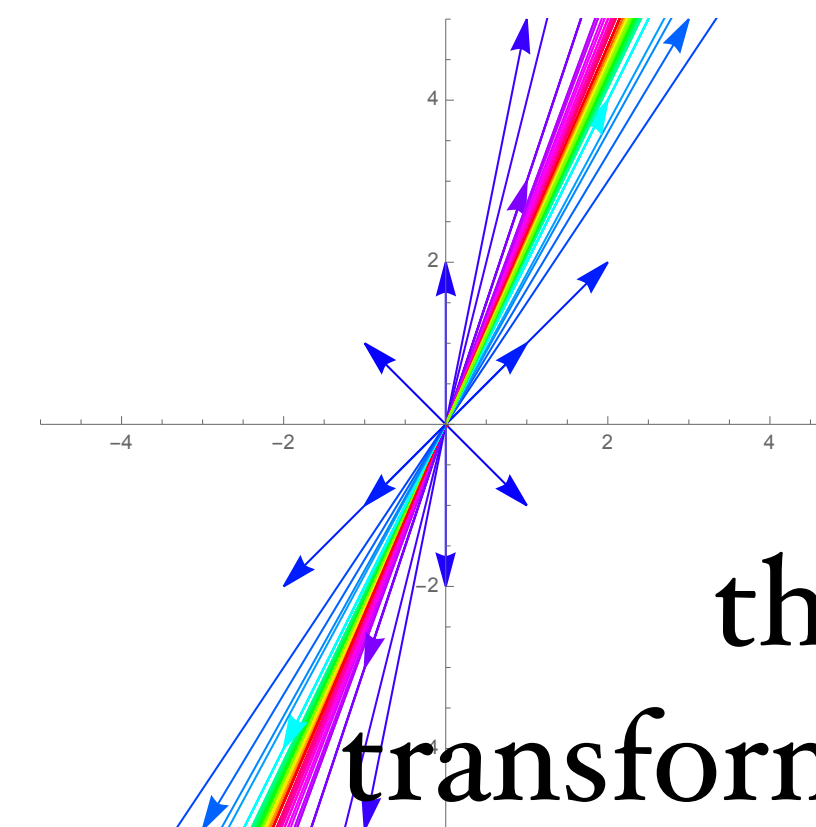
The solution is *unique* (if it exists) if:

matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \text{?}$$



$\downarrow \begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix}$



the linear transformation is 1-to-1

Next up:

- This week: Chapter 2.4-8 of *LADW*
- Quiz 3 at end of class today.
- Homework 2 is due today / tomorrow.
- Problem session today 3-5pm in Ryerson 177. (Possibly snacks.)
- ***Change:*** Open office hours on Sunday 3-4pm in **Cobb 115**
- Homework 3 distributed today, due end of next week.
- First problem swap problems due next Wednesday.